

Under-five mortality associated with antimicrobial-resistant bacteria in Ethiopia

Background

- The high child mortality burdens in resource-limited settings are attributable to various causes including infectious diseases, malnutrition, and congenital and birth defects.
- Resistance to broad-spectrum antimicrobials is probably the major cause of death from treatable bacterial infections after hospitalization in developing countries including Ethiopia. However, evidence are scarce on bacterial etiology contributing to under five deaths and their antimicrobial resistance (AMR) levels in Ethiopia.
- This study intends to address these evidence gaps using two novel causes of death studies conducted in Ethiopia: Child health and mortality prevention surveillance (CHAMPS) on 196 stillbirths and under-five deaths and Study of illness in preterm (SIP) on 1109 preterm deaths.

Key Findings

Bacterial pathogens in under-five deaths

- Sepsis and pneumonia accounted for 73.4% of immediate causes in under-five deaths; sepsis alone responsible for 61.2%.
- Klebsiella pneumoniae* (*K. pneumoniae*) was the leading with 42.8% of isolates followed by *Escherichia coli* (14.5%).
- Vaccine preventable pathogens (*Streptococcus pneumonia*, *Haemophilus influenza* and *Neisseria meningitidis*) accounted for 21.4% of isolates (Figure 1).

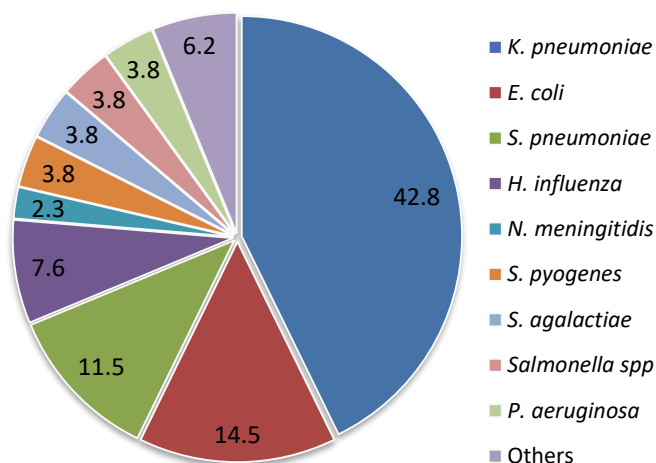


Fig. 1: Percentage of bacterial pathogens isolated from blood/CSF of dead children in Haramaya, Harar and Kersa HDSS, Ethiopia, 2019 to 2021. (n=131)

Bacterial pathogens in preterm deaths

- Bacterial infections (*sepsis, pneumonia and meningitis*) were the primary cause for 29.9% of preterm deaths; sepsis alone responsible for 26.1%.
- Seventeen bacterial species/groups with a total of 233 isolates were identified out of 1,109 preterm deaths.
- Of 17 bacterial species, *K. pneumonia*, coagulase negative staphylococcus (CoNS) and *Staphylococcus aureus* accounted 29.2%, 18% and 17.6% of isolates, respectively.
- Nearly half (47.5%) of isolates belong to the Enterobacteriace group mainly *E. coli*, *Klebsiella spp*, *Enterobacter*, *Acinetobacter* & *Serratia*.

Antimicrobial resistance of pathogens in under five deaths

- The two prevalent pathogens (*K. pneumonia* and *E. coli*) were wholly susceptible for Amikacin.
- But for the rest antimicrobials, the resistance of *K. pneumoniae* isolates ranged from 9.6% (Ciprofloxacin) to 100% (Penicillin) and >90% resistance to Ceftriaxone, Clindamycin, Ampicillin, Erythromycin and Gentamicin.
- The resistance of *E. coli* isolates ranged from 18.2% (Ciprofloxacin) to 100% (Clindamycin) with moderate to Amoxicillin-Clavuanate (25.9%), Gentamicin (29.6%) and Ceftriaxone (48.2%) while >60% for the rest of antimicrobials

Antimicrobial resistance of pathogens in preterm Infant

- The resistance of over 75% of *K. pneumoniae* isolates was >75% to all antimicrobials tested except Amikacin (12.5%), (Figure 2).
- S. aureus* isolates (n=9) were susceptible to vancomycin except one.

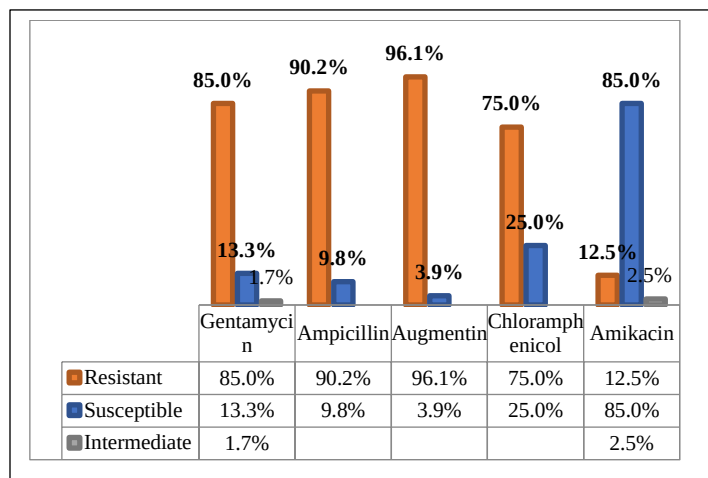
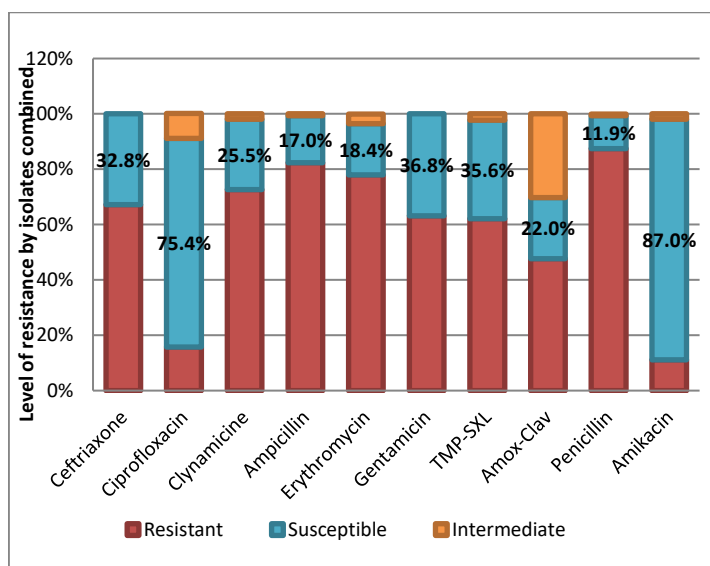


Figure 2: Antimicrobial resistance of *K. pneumonia* isolates in preterm infants in five referral hospitals of Ethiopia, 2016-2018

Susceptibility levels of antimicrobials in under five

- Of 10 antimicrobials tested, Amikacin and Ciprofloxacin had susceptibility levels of 87.0% and 75.4% against all isolates (n=161) in under-5 deaths, respectively.
- The rest had Susceptibility levels of less than 37% with the lowest (11.9%) for penicillin (Figure 3).



- Figure 3:** Susceptibility levels of antimicrobials in under-five deaths in Haramaya, Harar and Kersa HDSS, Ethiopia, 2019 to 2021

Susceptibility levels of antimicrobials in preterm Infant

- Of 9 antimicrobials with susceptibility testing results, Vancomycin and Amikacin had susceptibility levels of 81.8% and 73.3% against all isolates regardless of the pathogen. The rest had susceptibility level of below 50% with the lowest for Ampicillin (8.0%) and Penicillin G (6.2%).

Table 1: Susceptibility levels of antimicrobials in preterm infants in five referral hospitals of Ethiopia, 2016 to 2018

Anti-microbial	Total isolates with AST*	Susceptibility by all isolates (%)		
		Susceptible	Resistant	Intermediate
Vancomycin	33	27 (81.8)	6 (18.2)	-
Amikacin	86	63 (73.3)	21 (24.4)	2 (2.3)
Cloxacillin	28	13 (46.4)	15 (53.6)	-
Chloramphenicol	103	32 (31.1)	69 (67.0)	2 (1.9)
Gentamycin	150	22 (14.7)	127 (84.7)	1 (0.6)
Ceftazidime	58	7 (12.1)	51 (87.9)	-
Amox-clav	80	9 (11.3)	71 (88.7)	-
Ampicillin	100	8 (8.0)	92 (92.0)	-
Penicillin G	32	2 (6.2)	30 (93.8)	-

* AST: antimicrobial susceptibility testing

Conclusion and recommendation

- K. pneumonia* is the predominant cause of sepsis and pneumonia in child deaths followed by vaccine preventable, *Enterobacteriaceae* and *staphylococcus* bacteria groups.
- The unusual finding is that the pathogens are extensively resistant to the antimicrobials frequently used to treat sepsis and pneumonia in Ethiopia such as ceftriaxone, Augmentin and Gentamycin among others.
- The pathogens' greatest levels of antibiotic resistance indicate that the healthcare systems are run out of antimicrobial choices to treat life-threatening childhood infections.
- Hence, the healthcare system should look for antimicrobial options and strengthen antimicrobial stewardship, and consider new infection prevention strategies such as surveillance and isolation of child patients colonized by multidrug resistant pathogens during hospital admissions.

Acknowledgment

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